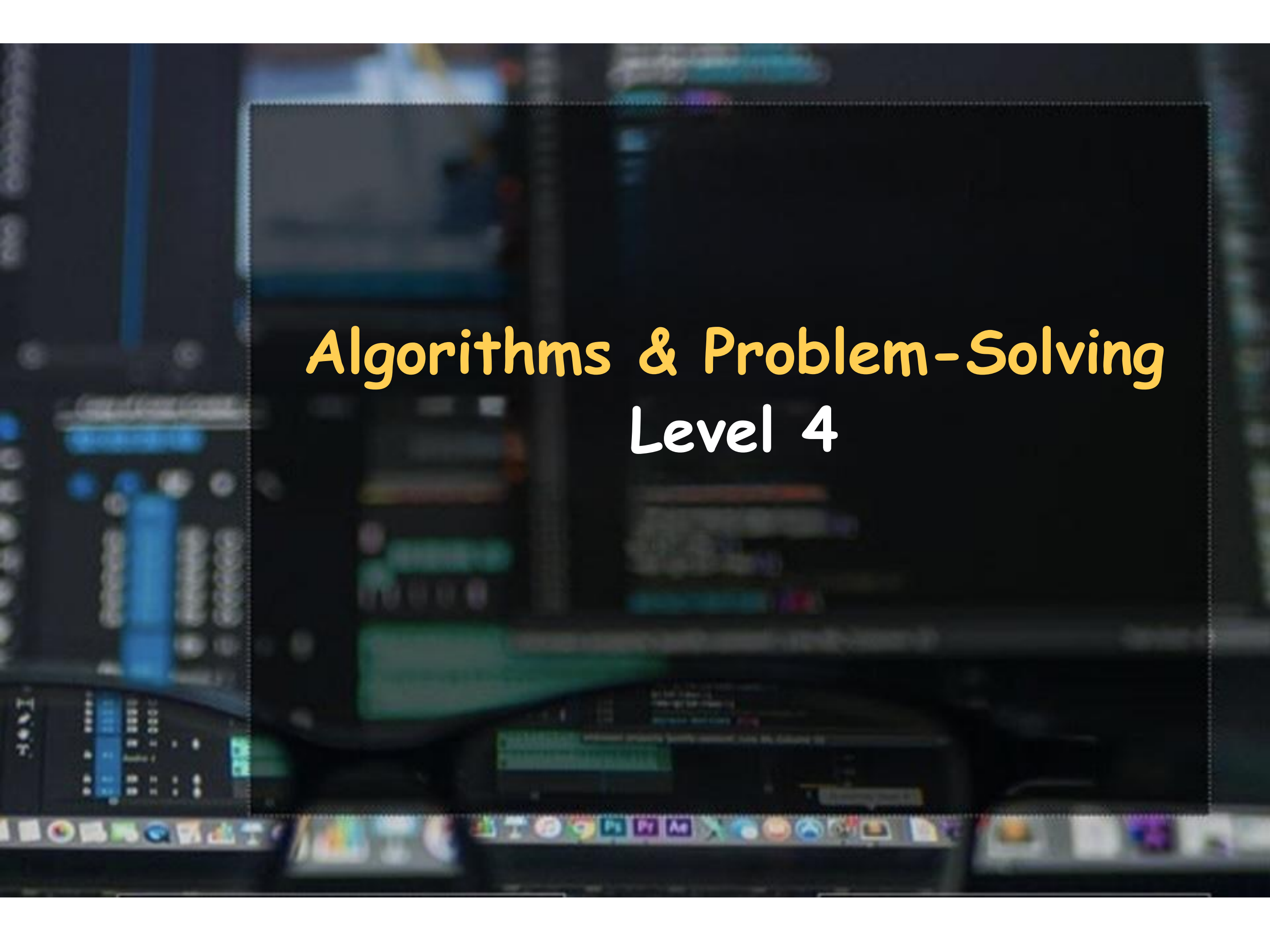
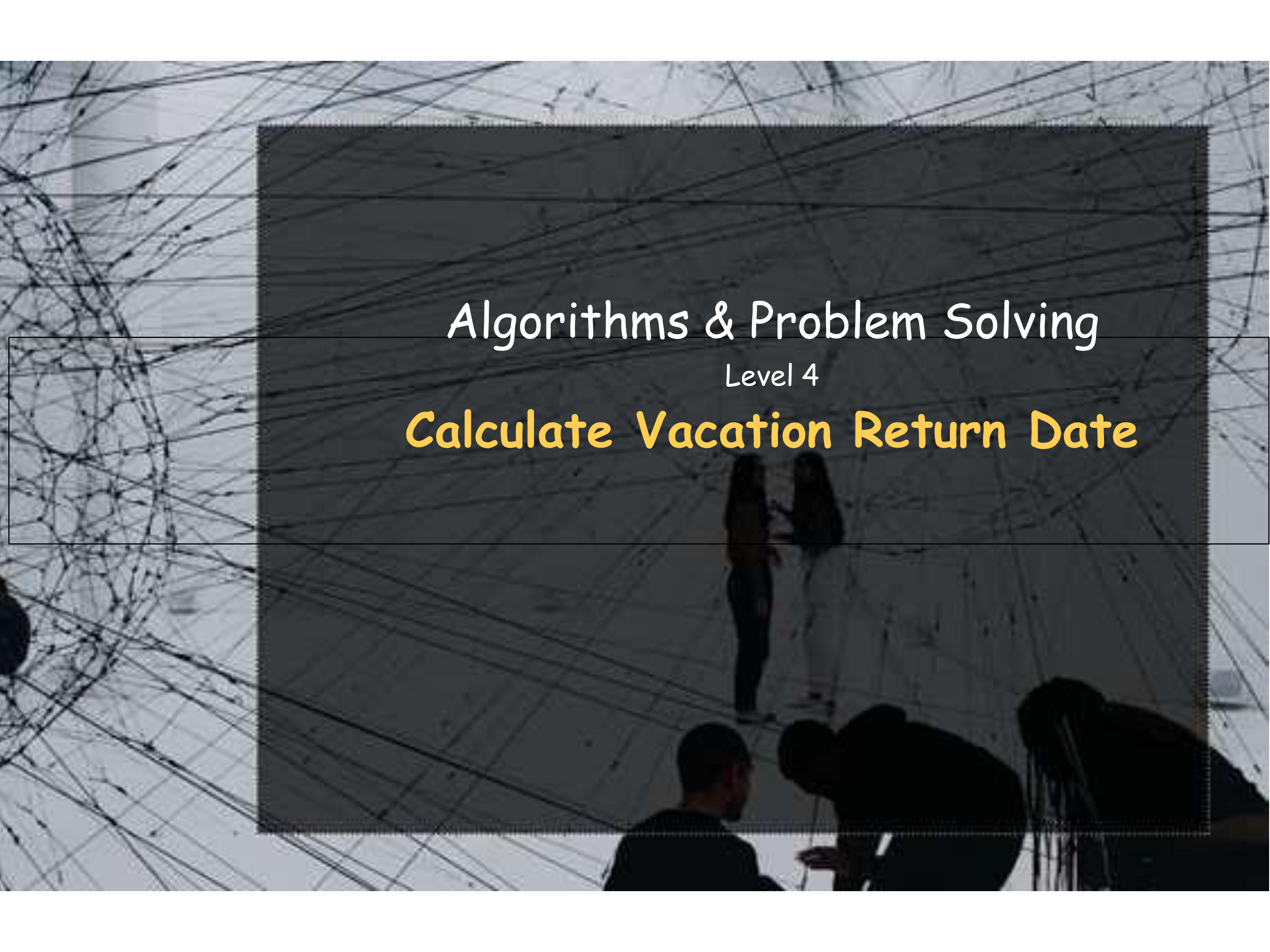




Algorithms & Problem-Solving Level 4

The background of the slide features a complex network diagram with numerous nodes and connecting lines, overlaid on a dark, semi-transparent rectangular area. In the lower portion of this area, the silhouettes of several people are visible, appearing to be in a collaborative work environment.

Algorithms & Problem Solving

Level 4

Calculate Vacation Return Date



Definition

- The goal is to create a C++ program that calculates the return date from vacation based on:
 - the vacation start date
 - and the number of vacation days, excluding weekends (Friday and Saturday) from the calculation.



Explanation:

- The program relies on the `sDate` struct to represent the date (day, month, year), and uses a set of helper functions to accurately process dates.

Procedures

- 1 Read vacation data from the user
 -  ReadFullDate()i To read the start date
 -  ReadVacationDays()i To read the number of vacation days
- 2 Check the validity of dates and years
 -  IsLeapYear(year)i Checks if it's a leap year
 -  NumberOfDaysInAMonth(month, year)i Displays the number of days in the month (taking leap years into account)
- 3 Advance the date day by day
 -  IncreaseDateByOneDay(date)i
 -  If we reach the end of the month, it advances to the next month
 -  If we reach the end of the year, it advances to the next year
- 4 Determine the day of the week
 -  DayOfWeekOrder(day, month, year)i Reorders the day of the week (0 for Sunday, 6 for Saturday)
 -  IsWeekEnd(date)i Checks if it's Friday Or Saturday
 -  IsBusinessDay(date)i Checks if today is a workday (Sunday to Thursday)

5 Calculating the Return Date

■ CalculateVacationReturnDate(dateFrom, vacationDays)i

■ Logical Steps:

- If the start date is a weekend → Ignores it and starts from the first workday
- Adds the number of vacation days, skipping holidays (Friday and Saturday)
- After the count is complete, if the return date falls on a weekend → Moves forward to the next workday



Example:

Vacation Starts:

Please enter a Day? 1

Please enter a Month? 1

Please enter a Year? 2022

Please enter vacation days? 23

Return Date: Wed , 2/2/2022



Explanation:

- The holiday begins on Saturday, January 1, 2022 (exempt holiday)
- It actually begins on Sunday, January 2, 2022
- After calculating 23 working days (excluding Fridays and Saturdays), the return date is Wednesday, February 2, 2022



Note:

- The program ignores all weekends before and during the holiday.
- The timer only includes actual working days.
- The holiday actually begins on the first working day after the start date if it falls on a holiday.

Summary

- ✓ The program uses:
 - Loops and conditional functions for accurate date handling
 - Real-world date logic with support for switching between months and years
 - Multiple functions for code reuse and organization

Thank you 😊